DATACO. SUPPLY CHAIN ANALYSIS

TOOL USED - EXCEL

MADE BY-

OISHIKA BOSE

PROBLEM STATEMENT

- The business faces **delivery delays, late shipments, and inconsistent profitability**.

- There is a need to understand **how shipping performance impacts profit and customer satisfaction**.

- Management lacks clarity on **which regions, products, and shipping modes drive delays or losses**.

- Identifying **high-risk orders and loss-making transactions** is critical for operational improvement.

OBJECTIVES

- Analyze **sales and profit performance across regions, products, and customer segments**.

- Identify **late deliveries and their impact on profit and order status**.

- Evaluate **shipping modes and delivery delays** to improve logistics efficiency.

- Detect **loss-making orders and high-risk delivery patterns**.

- Support **data-driven decisions** to improve delivery performance and profitability.

DATASET DESCRIPTION

- The dataset contains **order-level e-commerce transaction data**.
- Covers **sales, profit, discounts, shipping performance, delivery status, and customer details**.
- Includes **time-based fields** (order date, shipping date, Month-Year trend).
- Captures **geographic information** (country, region, state, city, market).
- Tracks **logistics performance** such as shipping mode, delivery delay, late delivery risk.
- Contains **financial metrics** like sales, profit per order, benefit per order, and profit flag.

DATA CLEANING

- Adjusted row and width sizes.
- Replaced null values using mode function.
- Few numeric column were in text format changed those into numeric format.
- Added helper columns for better analysis.
- Corrected date and time formats.
- Deleted few unwanted columns.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Type	Days for sh	Days for sh	Benefit per	Sales per c	Delivery St	Late_delive	Category Ic	Category N	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Departmer	Departmer	Latitude
2	DEBIT	3	4	91.25	314.64	Advance sl	0	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Cally	20755	Holloway	XXXXXXXX	Consumer	PR	5365 Noble	725	2	Fitness	18.25145
3	TRANSFER	5	4	-249.09	311.36	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Irene	19492	Luna	XXXXXXXX	Consumer	PR	2679 Rusti	725	2	Fitness	18.27945
4	CASH	4	4	-247.78	309.72	Shipping o	0	73	Sporting G	San Jose	EE. UU.	XXXXXXXX	Gillian	19491	Maldonado	XXXXXXXX	Consumer	CA	8510 Roun	95125	2	Fitness	37.29223
5	DEBIT	3	4	22.86	304.81	Advance sl	0	73	Sporting G	Los Angele	EE. UU.	XXXXXXXX	Tana	19490	Tate	XXXXXXXX	Home Offi	CA	3200 Ambe	90027	2	Fitness	34.12595
6	PAYMENT	2	4	134.21	298.25	Advance sl	0	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Orli	19489	Hendricks	XXXXXXXX	Corporate	PR	8671 Iron A	725	2	Fitness	18.25377
7	TRANSFER	6	4	18.58	294.98	Shipping c	0	73	Sporting G	Tonawanda	EE. UU.	XXXXXXXX	Kimberly	19488	Flowers	XXXXXXXX	Consumer	NY	2122 Hazy	14150	2	Fitness	43.01397
8	DEBIT	2	1	95.18	288.42	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Constance	19487	Terrell	XXXXXXXX	Home Offi	PR	1879 Gree	725	2	Fitness	18.24254
9	TRANSFER	2	1	68.43	285.14	Late delive	1	73	Sporting G	Miami	EE. UU.	XXXXXXXX	Erica	19486	Stevens	XXXXXXXX	Corporate	FL	7595 Cotto	33162	2	Fitness	25.92887
10	CASH	3	2	133.72	278.59	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Nichole	19485	Olsen	XXXXXXXX	Corporate	PR	2051 Dusty	725	2	Fitness	18.23322
11	CASH	2	1	132.15	275.31	Late delive	1	73	Sporting G	San Ramor	EE. UU.	XXXXXXXX	Oprah	19484	Delacruz	XXXXXXXX	Corporate	CA	9139 Blue	94583	2	Fitness	37.77399
12	TRANSFER	6	2	130.58	272.03	Shipping c	0	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Germane	19483	Short	XXXXXXXX	Corporate	PR	4058 Quiet	725	2	Fitness	18.28284
13	TRANSFER	5	2	45.69	268.76	Late delive	1	73	Sporting G	Freeport	EE. UU.	XXXXXXXX	Freya	19482	Robbins	XXXXXXXX	Consumer	NY	3243 Shad	11520	2	Fitness	40.65487
14	TRANSFER	4	2	21.76	262.2	Late delive	1	73	Sporting G	Salinas	EE. UU.	XXXXXXXX	Cassandra	19481	Jensen	XXXXXXXX	Corporate	CA	131 Sunny	93905	2	Fitness	36.67634
15	DEBIT	2	1	24.58	245.81	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Natalie	19480	Mcfadden	XXXXXXXX	Corporate	PR	2531 Wish	725	2	Fitness	18.27844
16	TRANSFER	2	1	16.39	327.75	Late delive	1	73	Sporting G	Peabody	EE. UU.	XXXXXXXX	Kimberley	19479	Sharpe	XXXXXXXX	Corporate	MA	6417 Silver	1960	2	Fitness	42.52628
17	DEBIT	2	1	-259.58	324.47	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Sade	19478	Lancaster	XXXXXXXX	Corporate	PR	257 Harves	725	2	Fitness	18.28405
18	PAYMENT	5	2	-246.36	321.2	Late delive	1	73	Sporting G	Canovanas	Puerto Ric	XXXXXXXX	Brynne	19477	Giles	XXXXXXXX	Corporate	PR	7342 Hazy	729	2	Fitness	18.39579
19	CASH	2	1	23.84	317.92	Late delive	1	73	Sporting G	Paramoun	EE. UU.	XXXXXXXX	Ciara	19476	Bird	XXXXXXXX	Corporate	CA	7787 Lazy	90723	2	Fitness	33.89869
20	DEBIT	2	1	102.26	314.64	Late delive	1	73	Sporting G	Caguas	Puerto Ric	XXXXXXXX	Bo	19475	Griffin	XXXXXXXX	Consumer	PR	5136 Rusti	725	2	Fitness	18.23378
21	PAYMENT	0	0	87.18	311.36	Shipping o	0	73	Sporting G	Mount Pro	EE. UU.	XXXXXXXX	Kim	19474	Simon	XXXXXXXX	Consumer	IL	1723 Tawn	60056	2	Fitness	42.04831

Type	Days for shipping (real	Days for shipment (scheduled	Benefit per orde	Sales per customer	Delivery Status	Late_delivery_ris	Category Id	Category Name	Customer City	Customer Country	Customer Id	Customer Se
DEBIT	3	4	91.25	314.64	Advance shipping	0	73	Sporting Goods	Caguas	Puerto Rico	20755	Consumer
TRANSFER	5	4	-249.09	311.36	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	19492	Consumer
CASH	4	4	-247.78	309.72	Shipping on time	0	73	Sporting Goods	San Jose	EE. UU.	19491	Consumer
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TRANSFER	6	4	18.58	294.98	Shipping canceled	0	73	Sporting Goods	Tonawanda	EE. UU.	19488	Consumer
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TRANSFER	2	1	68.43	285.14	Late delivery	1	73	Sporting Goods	Miami	EE. UU.	19486	Corporate
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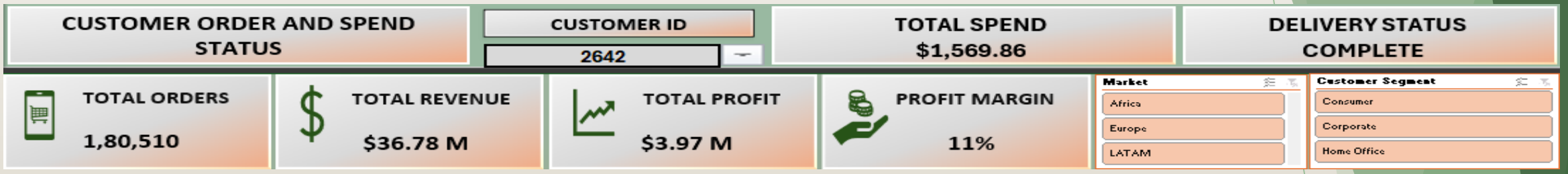
DEBIT	2	1	-259.58	324.47	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	19478	Corporate

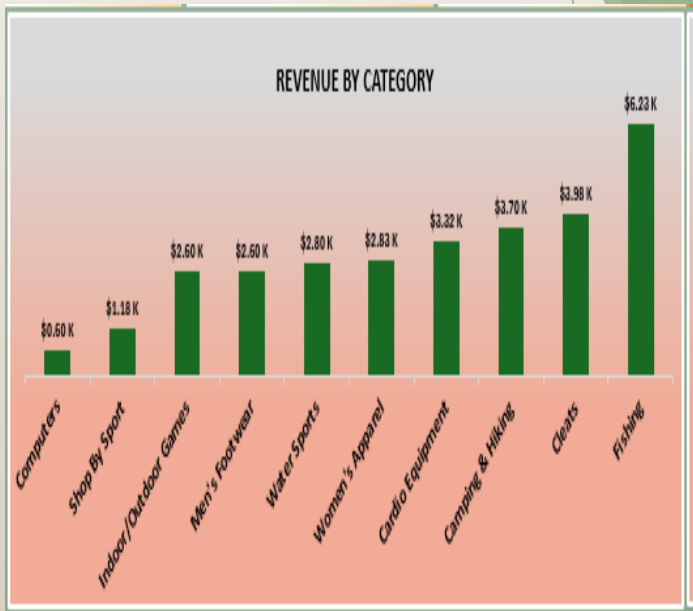
KEY METRICS AND ADVANCED METHODS USED

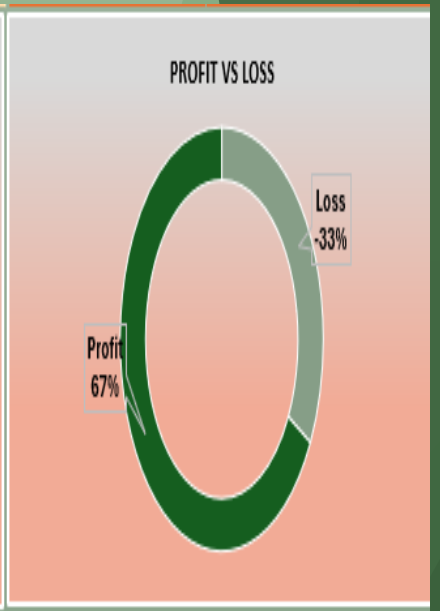


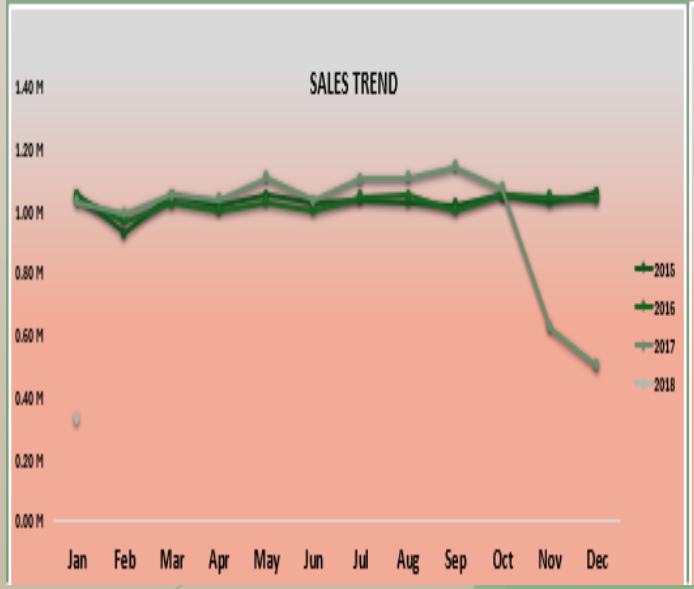
- **Customer order and spend status:** I have used data validation for fetching the **customer IDs.** and used **sumifs** to fetch the **total spend** linking with the customer Id using cell reference.
- **Delivery Status-** Tells the delivery status of the customer. I have used **xlookup** function to fetch the data using the customer Id and matching it from the dataset.
- **Total Orders**: The business has processed a total of 1,80,510 orders.
- **Total Revenue**: The cumulative sales generated from these orders amount to $36.78 million.
- **Total Profit**: After accounting for costs, the total earnings are $3.97 million.
- **Profit Margin**: The overall efficiency of the business is represented by a 11% profit margin.

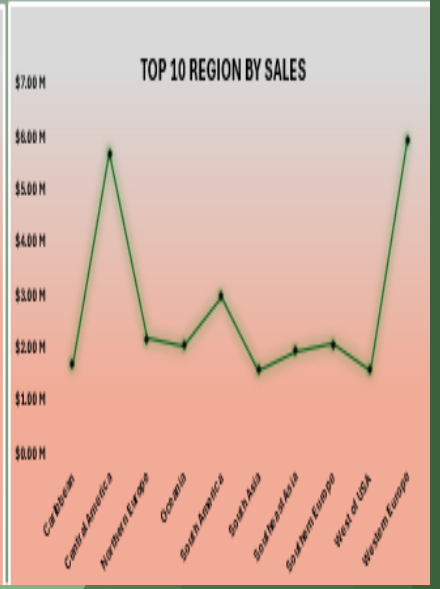
SALES & FINANCIAL PERFORMANCE

	OBSERVATION	KEY INSIGHTS
Revenue by Category	**Fishing** is the highest revenue-generating category at **$6.23 K**, followed by Cleats.	Sports and outdoor activities (Fishing, Camping, Hiking) are the primary revenue drivers for the business.
Sales Trend (2015-2018)	Sales were stable through 2017 but saw a **drastic decline in Q4 2017** and early 2018.	The sharp drop at the end of 2017 indicates a major market shift, supply chain issue, or data reporting gap that needs immediate investigation.
Profit vs Loss (Donut)	**67%** of transactions are profitable, while **33%** result in a loss.	A third of the business is losing money; management should identify if these losses are due to heavy discounting or high shipping costs.
Top 10 Region by Sales	**Western Europe** and **Central America** are the strongest performing regions.	Regional success is polarized; focus marketing efforts on Western Europe while investigating underperformance in Africa and parts of USA.









SALES & FINANCIAL PERFORMANCE

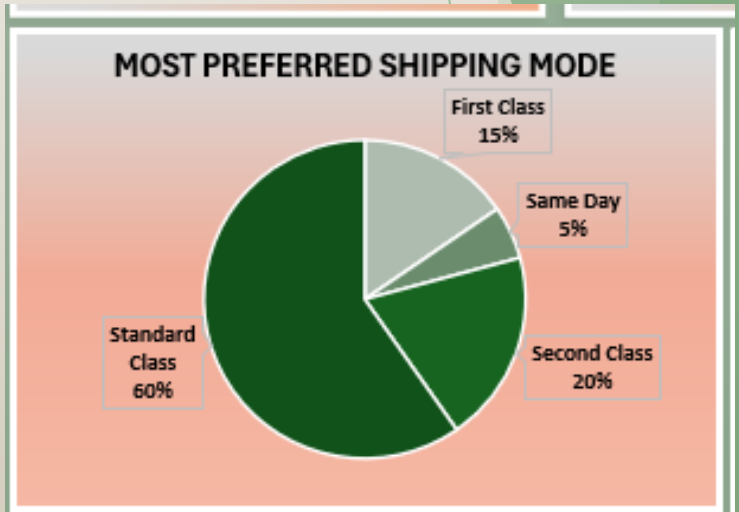
CHART	OBSERVATION	KEY INSIGHTS
Top 10 Products by Sales	The **"Field & Stream Sportsman 16 Gun Fire Safe"** is the leading product at **$6.93 K**.	Sales are concentrated in outdoor and safety gear; high-value items like the Fire Safe drive significant top-line revenue.

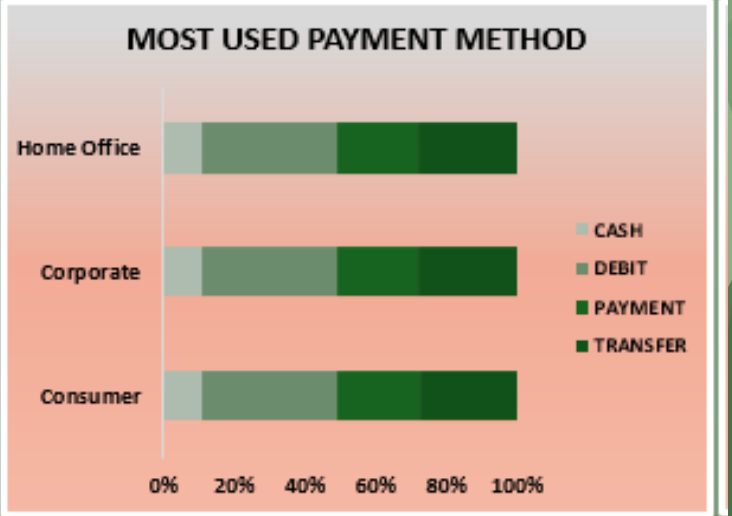
LOGISTICS & OPERATIONAL EFFICIENCY

METRIC	OBSERVATION	KEY INSIGHTS
% Late Delivery	More than half of the orders are delivered late	Indicates serious inefficiencies in the delivery process that may impact customer satisfaction
Average Delivery Delay	Delays are generally short but frequent	Even small delays, when consistent, can reduce customer trust and repeat purchases
Average Days for Shipping	Standard shipping duration is moderate	Improving logistics planning could reduce delays without increasing shipping time

LOGISTICS & OPERATIONAL EFFICIENCY

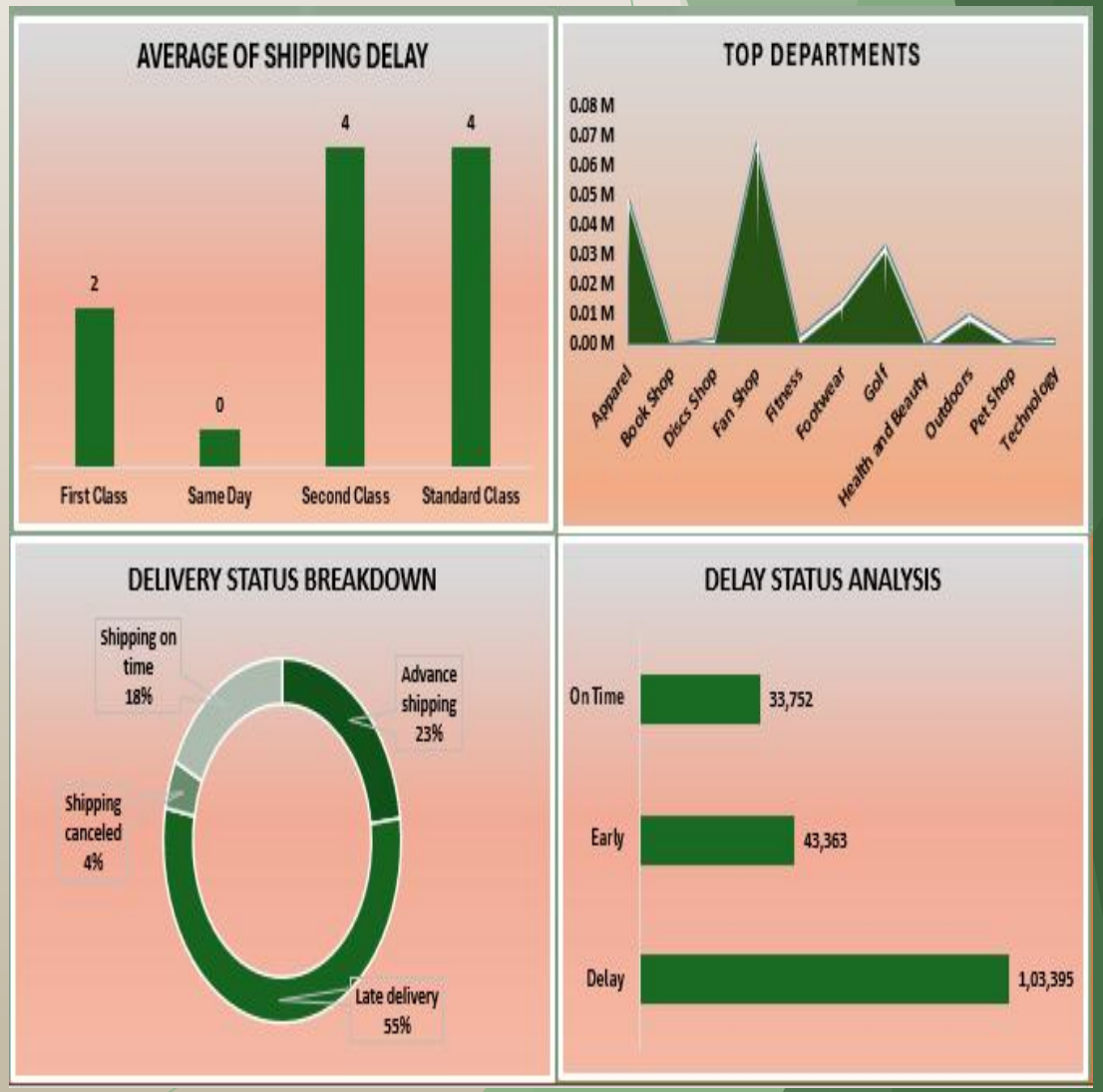
CHART	KEY OBSERVATION	INSIGHT
Most Preferred Shipping Mode	**Standard Class (60%)** is the most commonly used shipping mode	Customers prioritize **cost-effective shipping** over faster delivery options
	Second Class (20%) is the second preference	A segment of customers is willing to trade slightly higher cost for faster delivery
	Same Day (5%) has the lowest usage	High cost or limited availability may be restricting adoption
Most Used Payment Method	**Debit and Transfer** dominate across all customer segments	Indicates preference for **secure and non-cash digital payments**
	Consumer segment shows the highest overall payment activity	Consumers contribute the most to transaction volume



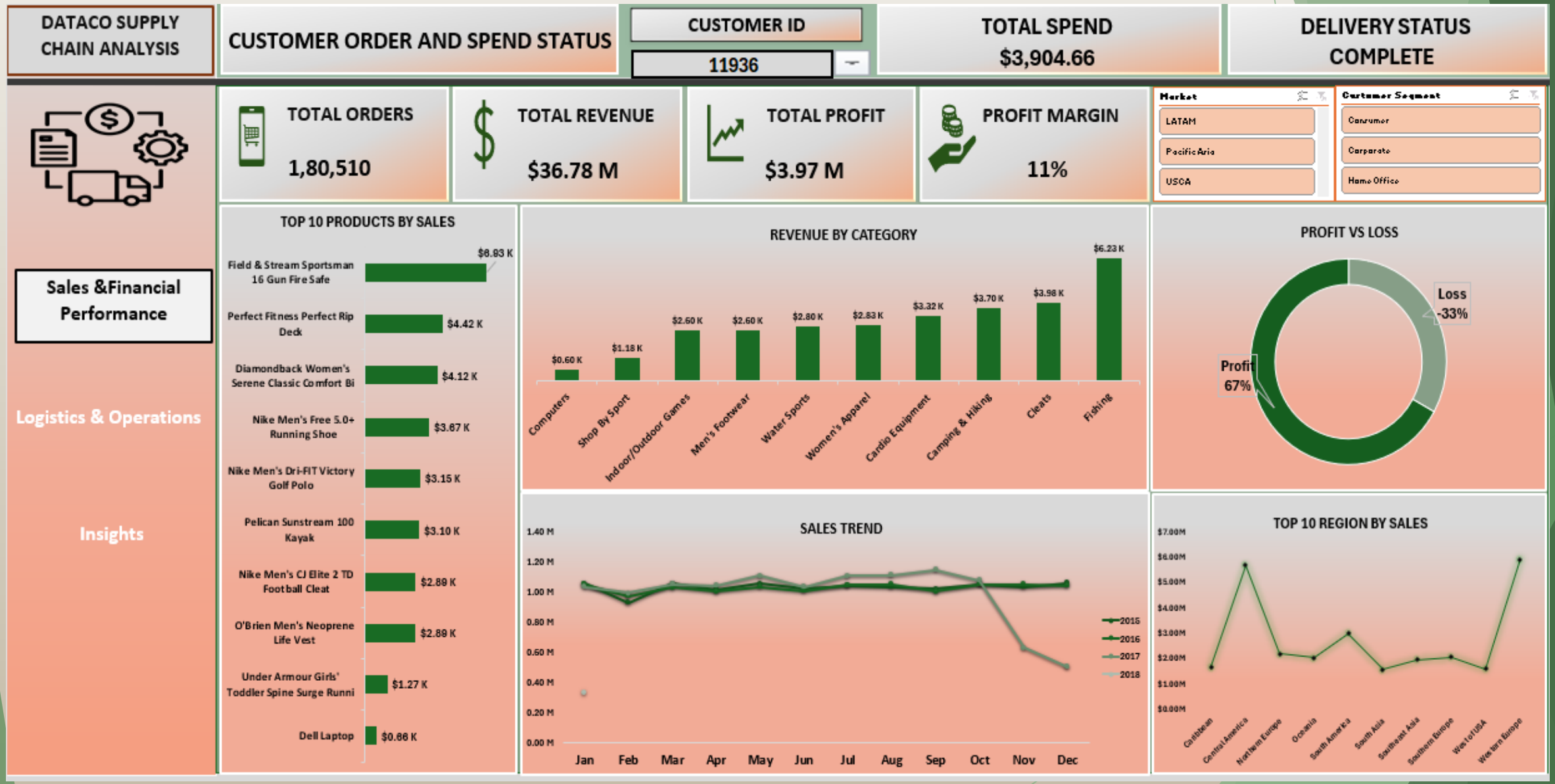


LOGISTICS & OPERATIONAL EFFICIENCY

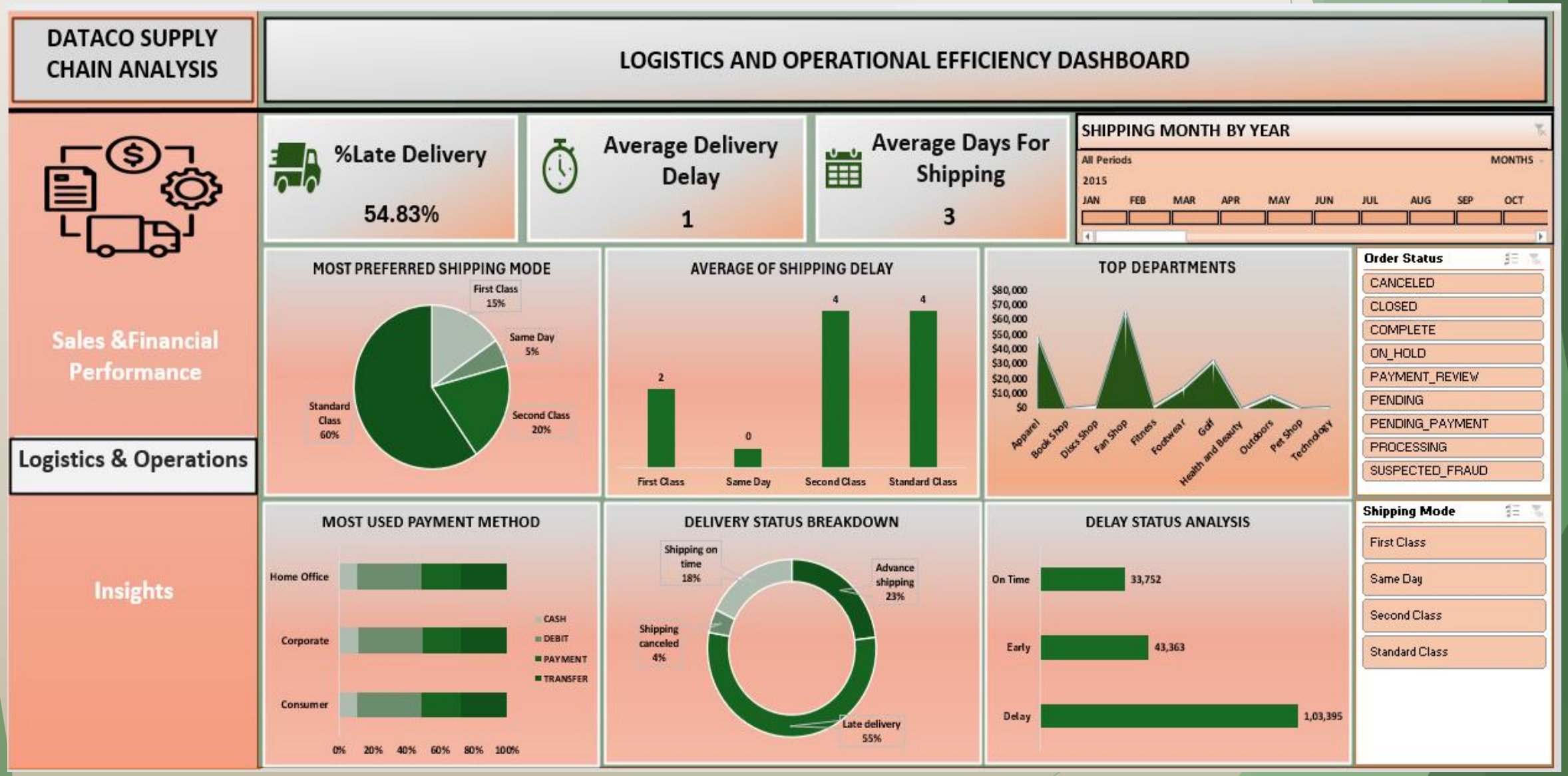
CHART	OBSERVATION	KEY INSIGHT
Average of Shipping Delay	Standard and Second Class shipments show the highest average delay (4 days)	Lower-cost shipping modes significantly impact delivery timelines
Top Departments	Fan Shop leads in sales contribution among all departments	Sports merchandise is the primary revenue-generating category
Delivery Status Breakdown	Late deliveries account for the majority (55%) of total shipments	Delivery performance issues are widespread and systemic
Delay Status Analysis	Delayed orders far exceed early and on-time deliveries	Logistics inefficiencies are the main cause of poor delivery reliability



DASHBOARD


DATACO SUPPLY CHAIN ANALYSIS
CUSTOMER ORDER AND SPEND STATUS
CUSTOMER ID
11936
TOTAL SPEND
$3,904.66
DELIVERY STATUS
COMPLETE
TOTAL ORDERS
1,80,510
TOTAL REVENUE
$36.78 M
TOTAL PROFIT
$3.97 M
PROFIT MARGIN
11%
Market
LATAM
Pacific Aria
USCA
Customer Segment
Consumer
Corporate
Home Office
Sales &Financial Performance
Logistics & Operations
Insights
TOP 10 PRODUCTS BY SALES
Field & Stream Sportsman 16 Gun Fire Safe $8.93 K
Perfect Fitness Perfect Rip Deck $4.42 K
Diamondback Women's Serene Classic Comfort Bi $4.12 K
Nike Men's Free 5.0+ Running Shoe $3.87 K
Nike Men's Dri-FIT Victory Golf Polo $3.15 K
Pelican Sunstream 100 Kayak $3.10 K
Nike Men's CJ Elite 2 TD Football Cleat $2.89 K
O'Brien Men's Neoprene Life Vest $2.89 K
Under Armour Girls' Toddler Spine Surge Runni $1.27 K
Dell Laptop $0.66 K
REVENUE BY CATEGORY
Computers $0.60 K
Shop By Sport $1.18 K
Indoor/Outdoor Games $2.60 K
Men's Footwear $2.60 K
Water Sports $2.80 K
Women's Apparel $2.83 K
Cardio Equipment $3.32 K
Camping & Hiking $3.70 K
Cleats $3.98 K
Fishing $6.23 K
PROFIT VS LOSS
Loss -33%
Profit 67%
SALES TREND
1.40 M
1.20 M
1.00 M
0.80 M
0.60 M
0.40 M
0.20 M
0.00 M
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2015
2016
2017
2018
TOP 10 REGION BY SALES
$7.00M
$6.00M
$5.00M
$4.00M
$3.00M
$2.00M
$1.00M
$0.00M
Caribbean
Central America
Northern Europe
Oceania
South America
South Asia
Southeast Asia
Southern Europe
Western USA
Western Europe

DASHBOARD


DATACO SUPPLY CHAIN ANALYSIS
LOGISTICS AND OPERATIONAL EFFICIENCY DASHBOARD
Sales &Financial Performance
Logistics & Operations
Insights
%Late Delivery
54.83%
Average Delivery Delay
1
Average Days For Shipping
3
SHIPPING MONTH BY YEAR
All Periods
MONTHS
2015
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT
MOST PREFERRED SHIPPING MODE
First Class 15%
Same Day 5%
Second Class 20%
Standard Class 60%
AVERAGE OF SHIPPING DELAY
2
0
4
4
First Class
Same Day
Second Class
Standard Class
TOP DEPARTMENTS
$80,000
$70,000
$60,000
$50,000
$40,000
$30,000
$20,000
$10,000
$0
Apparel
Book Shop
Discs Shop
Fan Shop
Fitness
Footwear
Golf
Health and Beauty
Outdoors
Pet Shop
Technology
Order Status
CANCELED
CLOSED
COMPLETE
ON_HOLD
PAYMENT_REVIEW
PENDING
PENDING_PAYMENT
PROCESSING
SUSPECTED_FRAUD
MOST USED PAYMENT METHOD
Home Office
Corporate
Consumer
CASH
DEBIT
PAYMENT
TRANSFER
0% 20% 40% 60% 80% 100%
DELIVERY STATUS BREAKDOWN
Shipping on time 18%
Advance shipping 23%
Shipping canceled 4%
Late delivery 55%
DELAY STATUS ANALYSIS
On Time
33,752
Early
43,363
Delay
1,03,395
Shipping Mode
First Class
Same Day
Second Class
Standard Class

SUMMERY, CONCLUSION & SUGGESTION

SUMMARY	Strong sales and revenue growth are driven by top products and key regions, but high delivery delays are impacting operational efficiency
CONCLUSION	The business is profitable, yet logistics inefficiencies—especially late deliveries—are limiting margin improvement and customer experience
SUGGESTION	Optimize shipping modes and delivery processes (especially Standard Class) to reduce delays and improve overall profitability and customer satisfaction.

THANK YOU